

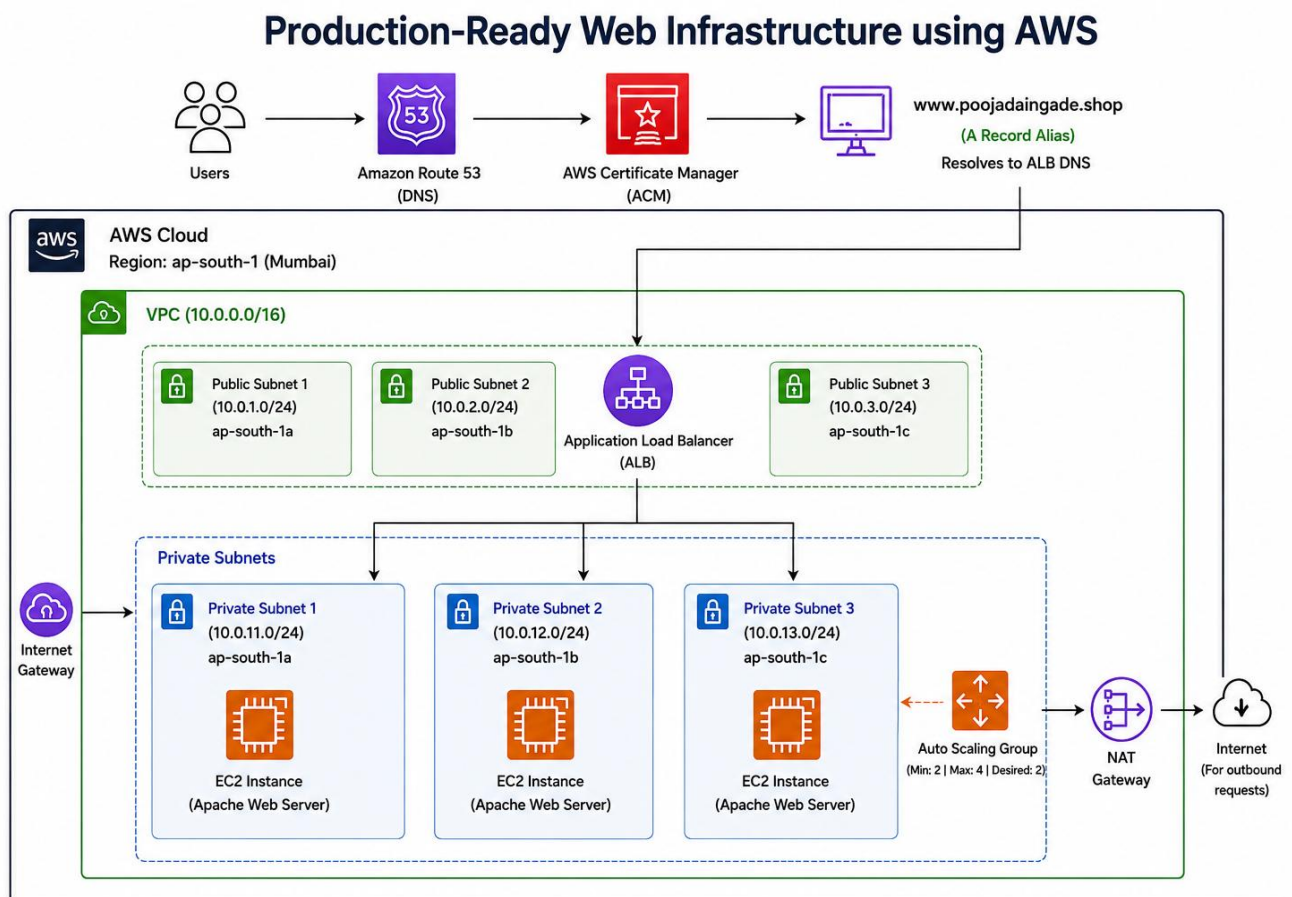
Production-Ready Web Infrastructure using AWS

Prepared By: Pooja Daingade

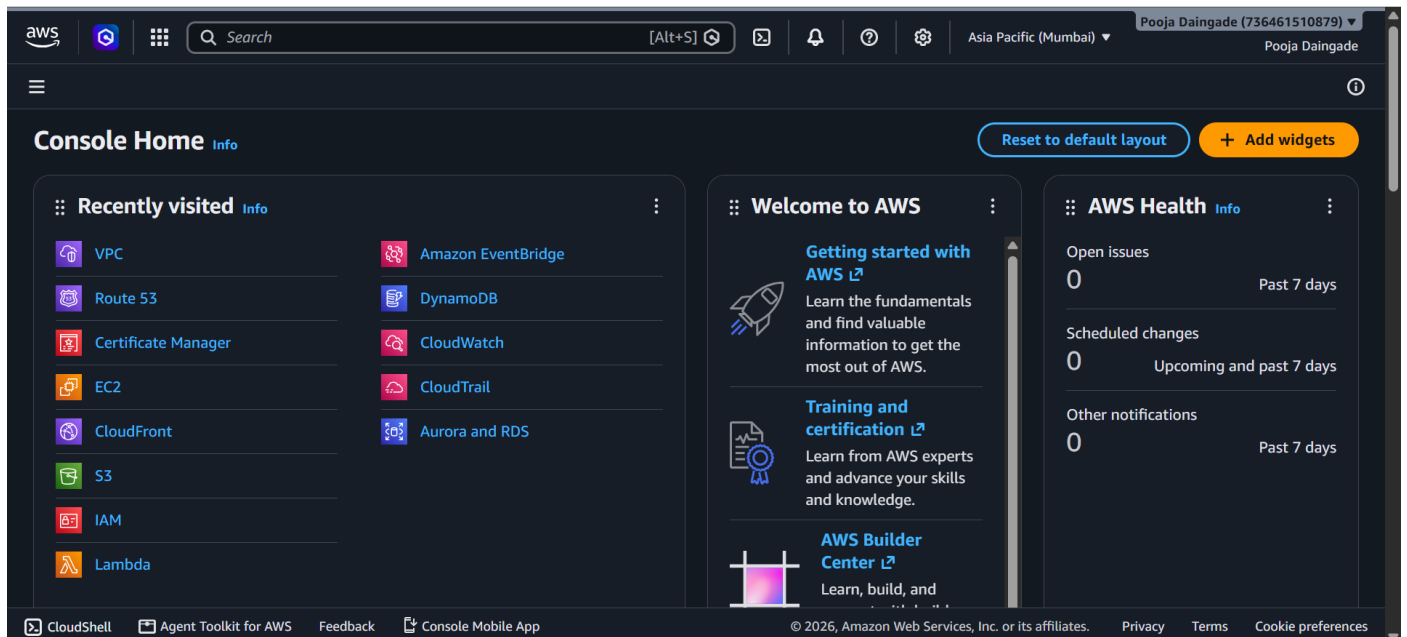
Project Overview

Designed and implemented a production-ready web server infrastructure on AWS to provide high availability, scalability, and reliable traffic distribution. The infrastructure uses a custom VPC with public and private subnets across three Availability Zones, along with Internet Gateway and NAT Gateways for network connectivity. Apache web servers are deployed on EC2 instances using a Launch Template and managed through an Auto Scaling Group. An Application Load Balancer distributes incoming traffic across the EC2 instances, while ACM and Route 53 are configured to provide HTTPS access through a custom domain.

Architecture Diagram

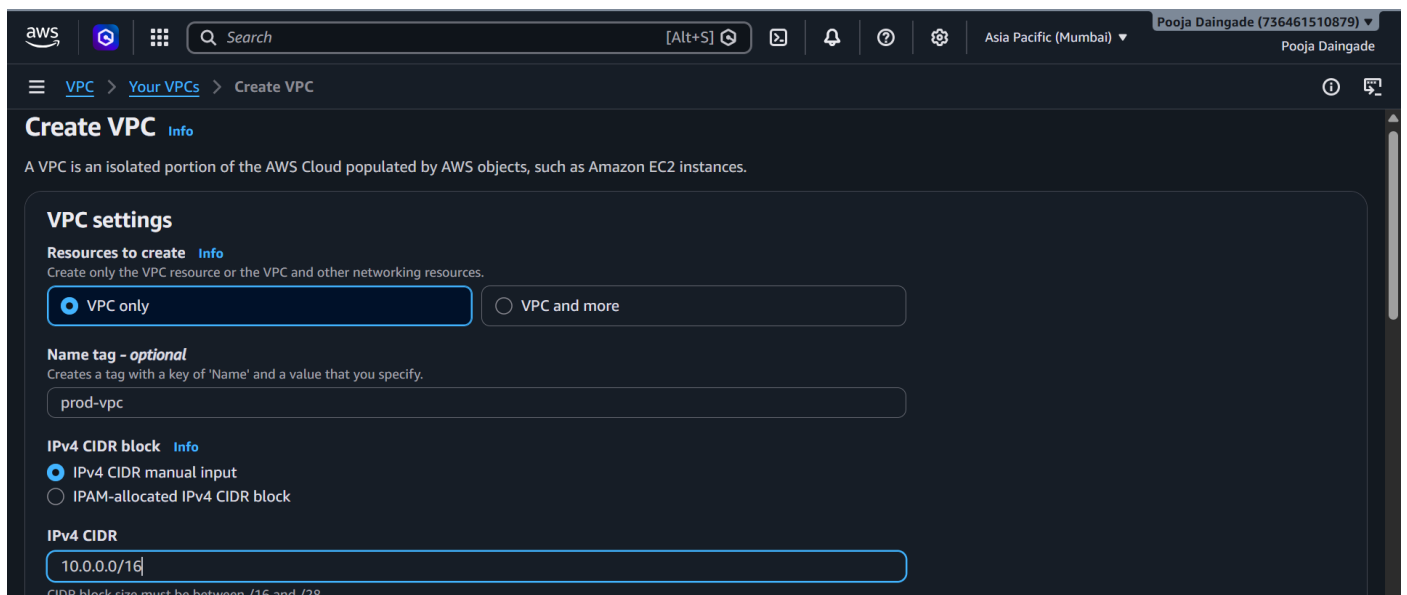


1. Logged in to the AWS Management Console and accessed the required AWS services for the project, including

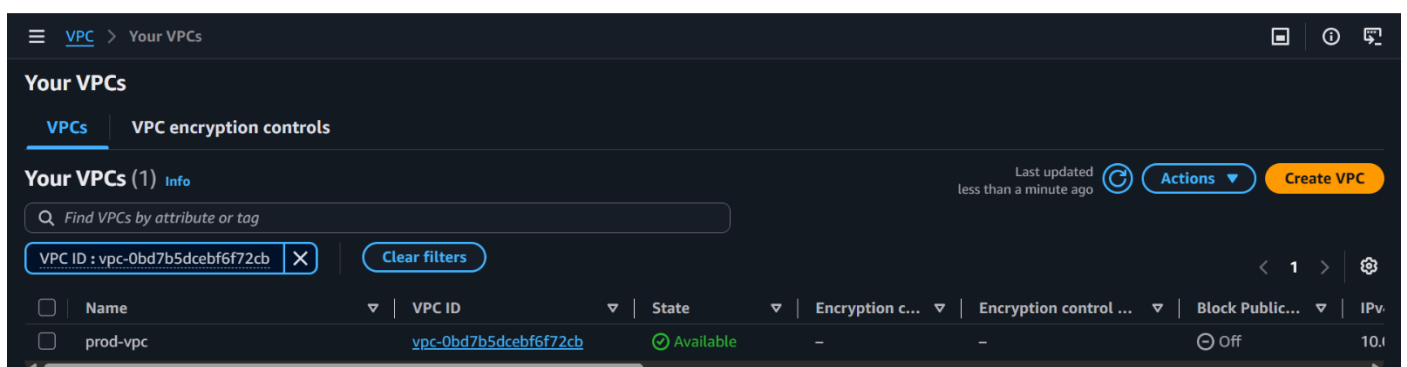


2. Create VPC

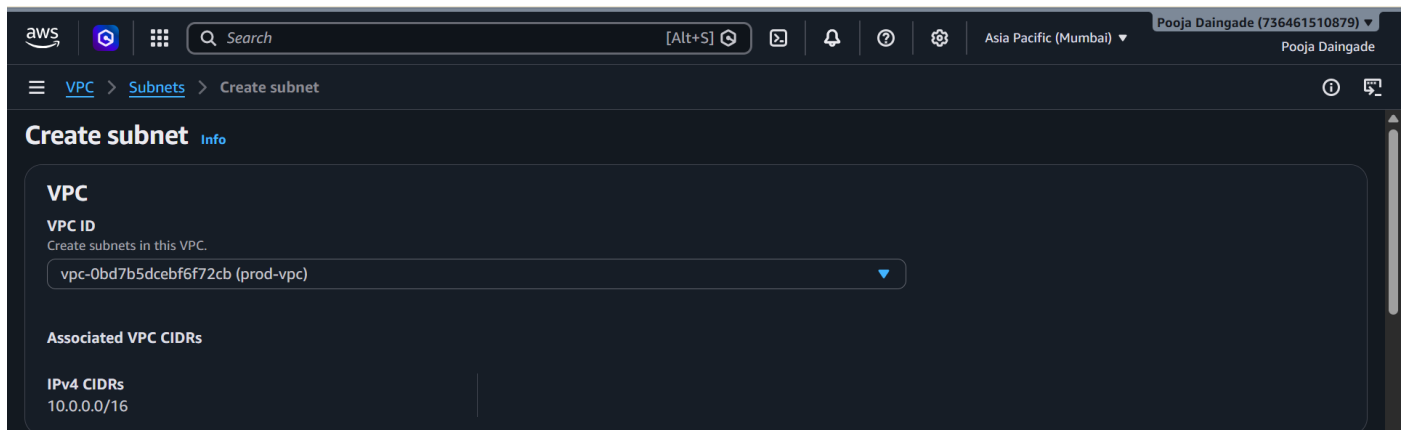
- VPC Name: prod-vpc
- IPv4 CIDR: 10.0.0.0/16



- VPC **prod-vpc** was successfully created.



3. Create Subnets: Select VPC ID : **prod-vpc**. (We create 3 public subnets across 3 AZs)



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VPC > Subnets > Create subnet

Create subnet [Info](#)

VPC

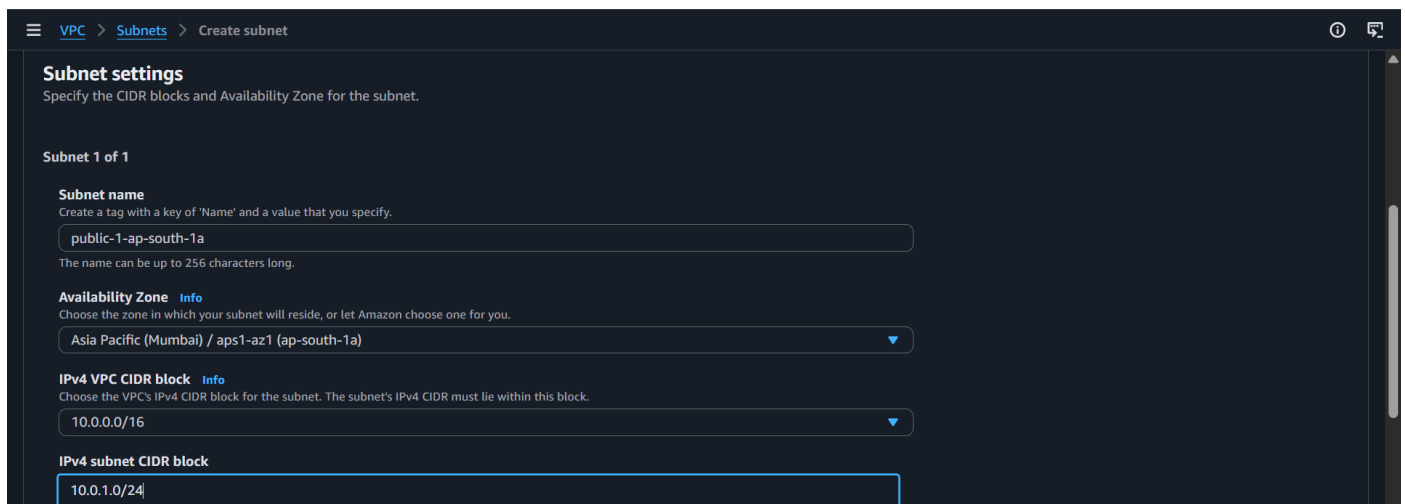
VPC ID
Create subnets in this VPC.

vpc-0bd7b5dcebf6f72cb (prod-vpc)

Associated VPC CIDRs

IPv4 CIDRs
10.0.0.0/16

- **Subnet 1** - Name: public-1-ap-south-1a
AZ : ap-south-1a
Subnet CIDR: 10.0.1.0/24



VPC > Subnets > Create subnet

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name
Create a tag with a key of 'Name' and a value that you specify.

public-1-ap-south-1a
The name can be up to 256 characters long.

Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.

Asia Pacific (Mumbai) / ap-s1-az1 (ap-south-1a)

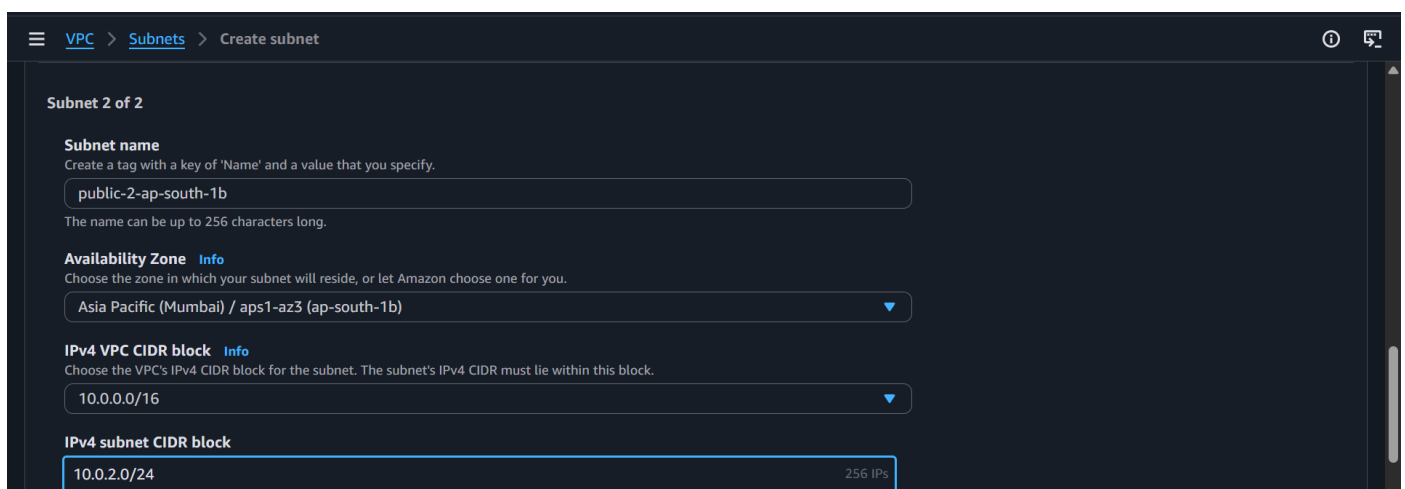
IPv4 VPC CIDR block [Info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

10.0.0.0/16

IPv4 subnet CIDR block

10.0.1.0/24

- **Subnet 2** - Name: public-2-ap-south-1b
AZ : ap-south-1b
Subnet CIDR: 10.0.2.0/24



VPC > Subnets > Create subnet

Subnet 2 of 2

Subnet name
Create a tag with a key of 'Name' and a value that you specify.

public-2-ap-south-1b
The name can be up to 256 characters long.

Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.

Asia Pacific (Mumbai) / ap-s1-az3 (ap-south-1b)

IPv4 VPC CIDR block [Info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

10.0.0.0/16

IPv4 subnet CIDR block

10.0.2.0/24 256 IPs

- **Subnet 3** - Name: public-3-ap-south-1c
AZ : ap-south-1c
Subnet CIDR: 10.0.3.0/24

The screenshot shows the 'Create subnet' page in the AWS Management Console. The breadcrumb navigation is 'VPC > Subnets > Create subnet'. The page title is 'Subnet 3 of 3'. The 'Subnet name' field is 'public-3-ap-south-1c'. The 'Availability Zone' is 'Asia Pacific (Mumbai) / ap-south-1c'. The 'IPv4 VPC CIDR block' is '10.0.0.0/16'. The 'IPv4 subnet CIDR block' is '10.0.3.0/24', with a note indicating it provides 256 IPs.

4. Create Subnets: Select VPC ID : **prod-vpc**. (We create 3 private subnets across 3 AZs)

The screenshot shows the 'Create subnet' page in the AWS Management Console. The breadcrumb navigation is 'VPC > Subnets > Create subnet'. The page title is 'Create subnet'. The 'VPC ID' is 'vpc-0bd7b5dcebf6f72cb (prod-vpc)'. The 'Associated VPC CIDRs' section shows 'IPv4 CIDRs' as '10.0.0.0/16'.

- **Subnet 1** - Name: private-1-ap-south-1a
AZ : ap-south-1a
Subnet CIDR: 10.0.11.0/24

The screenshot shows the 'Create subnet' page in the AWS Management Console. The breadcrumb navigation is 'VPC > Subnets > Create subnet'. The page title is 'Subnet settings'. The 'Subnet name' field is 'private-1-ap-south-1a'. The 'Availability Zone' is 'Asia Pacific (Mumbai) / ap-south-1a'. The 'IPv4 VPC CIDR block' is '10.0.0.0/16'. The 'IPv4 subnet CIDR block' is '10.0.11.0/24', with a note indicating it provides 256 IPs.

- **Subnet 2** - Name: private-2-ap-south-1b
AZ : ap-south-1b
Subnet CIDR: 10.0.12.0/24

Subnet 2 of 2

Subnet name
Create a tag with a key of 'Name' and a value that you specify.
private-2-ap-south-1b
The name can be up to 256 characters long.

Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
Asia Pacific (Mumbai) / ap-south-1b

IPv4 VPC CIDR block [Info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
10.0.0.0/16

IPv4 subnet CIDR block
10.0.12.0/24 256 IPs

- **Subnet 3** - Name: private-3-ap-south-1c
AZ : ap-south-1c
Subnet CIDR: 10.0.13.0/24

Subnet 3 of 3

Subnet name
Create a tag with a key of 'Name' and a value that you specify.
private-3-ap-south-1c
The name can be up to 256 characters long.

Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
Asia Pacific (Mumbai) / ap-south-1c

IPv4 VPC CIDR block [Info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
10.0.0.0/16

IPv4 subnet CIDR block
10.0.13.0/24

- Successfully created public and private subnets in the prod-vpc VPC.

Subnets (6) [Info](#)

Find subnets by attribute or tag

VPC = vpc-0bd7b5dceb6f72cb [Clear filters](#)

☐	Name	Subnet ID	State	VPC	Block Public...
☐	public-3-ap-south-1c	subnet-078365adcfac62a2d	Available	vpc-0bd7b5dceb6f72cb prod-...	Off
☐	private-1-ap-south-1a	subnet-0306d0466cb1b63e4	Available	vpc-0bd7b5dceb6f72cb prod-...	Off
☐	public-2-ap-south-1b	subnet-0f624c4b58ca6992b	Available	vpc-0bd7b5dceb6f72cb prod-...	Off
☐	private-2-ap-south-1b	subnet-0f3237927a585608e	Available	vpc-0bd7b5dceb6f72cb prod-...	Off
☐	private-3-ap-south-1c	subnet-091465837b61ba11e	Available	vpc-0bd7b5dceb6f72cb prod-...	Off
☐	public-1-ap-south-1a	subnet-044ddeb56719a58aa	Available	vpc-0bd7b5dceb6f72cb prod-...	Off

Select a subnet

5. Create Internet Gateway (IGW) - Name: **my-prod-igw**

Create internet gateway [Info](#)

An internet gateway is a virtual router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below.

Internet gateway settings

Name tag
Creates a tag with a key of 'Name' and a value that you specify.

my-prod-igw

Tags - optional
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key **Value - optional**

Q Name X Q my-prod-igw X Remove

Add new tag

You can add 49 more tags.

Cancel Create internet gateway

6. Internet Gateway **my-prod-igw** successfully created and attached it to the **prod-vpc** VPC.

Internet gateways (1/1) [Info](#)

Find internet gateways by attribute or tag

VPC ID : vpc-0bd7b5dcebf6f72cb X Clear filters

☒	Name	Internet gateway ID	State	VPC ID	Owner
☒	my-prod-igw	igw-0d713a0b0ec478389	Attached	vpc-0bd7b5dcebf6f72cb prod-vpc	73646

igw-0d713a0b0ec478389 / my-prod-igw

7. Create **Route Table** for public subnets - Name: **my-public-rt** Vpc: **prod-vpc**

Create route table [Info](#)

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

my-public-rt

VPC
The VPC to use for this route table.

vpc-074dcae9553661e11 (prod-vpc)

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key **Value - optional**

Q Name X Q my-public-rt X Remove

Add new tag

You can add 49 more tags.

Cancel Create route table

Add Route: 0.0.0.0/0 → Internet Gateway

Edit routes: rtb-0c672ef2db6ad294e / my-public-rt

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	CreateRouteTable
0.0.0.0/0	Internet Gateway	-	No	CreateRoute

Add route **Cancel** **Preview** **Save changes**

Associate Subnets:

- public-1-ap-south-1a
- public-2-ap-south-1b
- public-3-ap-south-1c

Edit subnet associations
Change which subnets are associated with this route table.

Available subnets (3/6)

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☒ public-3-ap-south-1c	subnet-078365adcfac62a2d	10.0.3.0/24	-	Main (rtb-0641889a33fb9cf1b)
☐ private-1-ap-south-1a	subnet-0306d0466cb1b6...	10.0.11.0/24	-	Main (rtb-0641889a33fb9cf1b)
☒ public-2-ap-south-1b	subnet-0f624c4b58ca6992b	10.0.2.0/24	-	Main (rtb-0641889a33fb9cf1b)
☐ private-2-ap-south-1b	subnet-0f3237927a5856...	10.0.12.0/24	-	Main (rtb-0641889a33fb9cf1b)
☐ private-3-ap-south-1c	subnet-091465837b61ba...	10.0.13.0/24	-	Main (rtb-0641889a33fb9cf1b)
☒ public-1-ap-south-1a	subnet-044dbeb56719a5...	10.0.1.0/24	-	Main (rtb-0641889a33fb9cf1b)

Selected subnets

subnet-078365adcfac62a2d / public-3-ap-south-1c X subnet-0f624c4b58ca6992b / public-2-ap-south-1b X subnet-044dbeb56719a58aa / public-1-ap-south-1a X

Cancel **Save associations**

8. Create NAT Gateways:

- **NAT 1** - Name: my-nat-ap-south-1a
Availability mode: Zonal
Subnet: public-1-ap-south-1a
Elastic-ip: Allocate

Create NAT gateway [Info](#)

A highly available, managed Network Address Translation (NAT) service that instances in private subnets can use to connect to services in other VPCs, on-premises networks, or the internet.

NAT gateway settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

The name can be up to 256 characters long.

Availability mode [Info](#)
Choose whether to deploy across all zones in the region or restrict to a single availability zone.

☐ **Regional - new**
Scales automatically across all regional AZs, simplifying management for multi AZ deployments.

☒ **Zonal**
Provides granular control within a specific availability zone, adhering to subnet level settings.

Subnet
Select a subnet in which to create the NAT gateway.

Connectivity type
Select a connectivity type for the NAT gateway.
☒ **Public**
☐ Private

Elastic IP allocation ID [Info](#)
Assign an Elastic IP address to the NAT gateway.
 [Allocate Elastic IP](#)

- **NAT 2** - Name: my-nat-ap-south-1b
Availability mode: Zonal
Subnet: public-2-ap-south-1b
Elastic-ip: Allocate

Create NAT gateway [Info](#)

A highly available, managed Network Address Translation (NAT) service that instances in private subnets can use to connect to services in other VPCs, on-premises networks, or the internet.

NAT gateway settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

The name can be up to 256 characters long.

Availability mode [Info](#)
Choose whether to deploy across all zones in the region or restrict to a single availability zone.

☐ **Regional - new**
Scales automatically across all regional AZs, simplifying management for multi AZ deployments.

☒ **Zonal**
Provides granular control within a specific availability zone, adhering to subnet level settings.

Subnet
Select a subnet in which to create the NAT gateway.

Connectivity type
Select a connectivity type for the NAT gateway.
☒ **Public**
☐ Private

Elastic IP allocation ID [Info](#)
Assign an Elastic IP address to the NAT gateway.
 [Allocate Elastic IP](#)

- **NAT 3** - Name: my-nat-ap-south-1c
Availability mode: Zonal
Subnet: public-3-ap-south-1c
Elastic-ip: Allocate

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VPC > NAT gateways > Create NAT gateway

Create NAT gateway [Info](#)

A highly available, managed Network Address Translation (NAT) service that instances in private subnets can use to connect to services in other VPCs, on-premises networks, or the internet.

NAT gateway settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

The name can be up to 256 characters long.

Availability mode [Info](#)
Choose whether to deploy across all zones in the region or restrict to a single availability zone.

☐ **Regional - new**
Scales automatically across all regional AZs, simplifying management for multi AZ deployments.
 ☒ **Zonal**
Provides granular control within a specific availability zone, adhering to subnet level settings.

Subnet
Select a subnet in which to create the NAT gateway.

Connectivity type
Select a connectivity type for the NAT gateway.

☒ **Public**
☐ **Private**

Elastic IP allocation ID [Info](#)
Assign an Elastic IP address to the NAT gateway.

9. Create Route Table for private subnets –

- Route Table 1** - Name: my-private-rt-ap-south-1a
Vpc: prod-vpc

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VPC > Route tables > Create route table

Create route table [Info](#)

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

VPC
The VPC to use for this route table.

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key	Value - optional	
Q Name	Q my-private-rt-ap-south-1a	Remove

You can add 49 more tags.

Add Route: 0.0.0.0/0 → Internet Gateway

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VPC > Route tables > rtb-063d55dae3086804f > Edit routes

Edit routes: rtb-063d55dae3086804f / my-private-rt-ap-south-1a

Destination	Target	Status	Propagated	Route Origin	
10.0.0.0/16	local	Active	No	CreateRouteTable	
Q 0.0.0.0/0	NAT Gateway	-	No	CreateRoute	Remove

Use: "nat-00ce650d044b5eda3"

Associate Subnet: private-1-ap-south-1a

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VPC

Route tables

rtb-063d55dae3086804f

Edit subnet associations

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (1/6)

Filter subnet associations

< 1 >

	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☐	public-3-ap-south-1c	subnet-078365adcfac62a2d	10.0.3.0/24	-	rtb-0c672ef2db6ad294e / my-public-rt
☒	private-1-ap-south-1a	subnet-0306d0466cb1b63e4	10.0.11.0/24	-	Main (rtb-0641889a33fb9cf1b)
☐	public-2-ap-south-1b	subnet-0f624c4b58ca6992b	10.0.2.0/24	-	rtb-0c672ef2db6ad294e / my-public-rt
☐	private-2-ap-south-1b	subnet-0f3237927a585608e	10.0.12.0/24	-	Main (rtb-0641889a33fb9cf1b)
☐	private-3-ap-south-1c	subnet-091465837b61ba11e	10.0.13.0/24	-	Main (rtb-0641889a33fb9cf1b)
☐	public-1-ap-south-1a	subnet-044dbeb56719a58aa	10.0.1.0/24	-	rtb-0c672ef2db6ad294e / my-public-rt

Selected subnets

subnet-0306d0466cb1b63e4 / private-1-ap-south-1a

Cancel

Save associations

- Route Table 2 - Name: my-private-rt-ap-south-1b
Vpc: prod-vpc

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VPC

Route tables

Create route table

Create route table

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional

Create a tag with a key of 'Name' and a value that you specify.

my-private-rt-ap-south-1b

VPC

The VPC to use for this route table.

vpc-0bd7b5dceb6f6f72cb (prod-vpc)

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Q Name

Value - optional

Q my-private-rt-ap-south-1b

Remove

Add new tag

You can add 49 more tags.

Cancel

Create route table

Add Route: 0.0.0.0/0 → Internet Gateway

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VPC

Route tables

rtb-0631f7d256d765bf4

Edit routes

Edit routes: rtb-0631f7d256d765bf4 / my-private-rt-ap-south-1b

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	CreateRouteTable
Q 0.0.0.0/0	NAT Gateway	-	No	CreateRoute
	nat-0e6e00971ac7d34b1			

Add route

Cancel

Preview

Save changes

Associate Subnet: private-2-ap-south-1b

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Search

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VPC

Route tables

rtb-0631f7d256d765bf4

Edit subnet associations

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (1/6)

Filter subnet associations

	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☐	public-3-ap-south-1c	subnet-078365adcfac62a2d	10.0.3.0/24	-	rtb-0c672ef2db6ad294e / my-public-rt
☐	private-1-ap-south-1a	subnet-0306d0466cb1b63e4	10.0.11.0/24	-	rtb-063d55dae3086804f / my-private-rt-ap-south-1a
☐	public-2-ap-south-1b	subnet-0f624c4b58ca6992b	10.0.2.0/24	-	rtb-0c672ef2db6ad294e / my-public-rt
☒	private-2-ap-south-1b	subnet-0f3237927a585608e	10.0.12.0/24	-	Main (rtb-0641889a33fb9cf1b)
☐	private-3-ap-south-1c	subnet-091465837b61ba11e	10.0.13.0/24	-	Main (rtb-0641889a33fb9cf1b)
☐	public-1-ap-south-1a	subnet-044dbeb56719a58aa	10.0.1.0/24	-	rtb-0c672ef2db6ad294e / my-public-rt

Selected subnets

subnet-0f3237927a585608e / private-2-ap-south-1b

Cancel

Save associations

- Route Table 3 - Name: my-private-rt-ap-south-1c
Vpc: prod-vpc

aws

Search

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VPC

Route tables

Create route table

Create route table

A route table specifies how packets are forwarded between the subnets within your VPC, the Internet, and your VPN connection.

Route table settings

Name - optional

Create a tag with a key of 'Name' and a value that you specify.

my-private-rt-ap-south-1c

VPC

The VPC to use for this route table.

vpc-0bd7b5dceb6f672cb (prod-vpc)

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Q Name

Value - optional

Q my-private-rt-ap-south-1c

Remove

Add new tag

You can add 49 more tags.

Cancel

Create route table

Add Route: 0.0.0.0/0 → Internet Gateway

aws

Search

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VPC

Route tables

rtb-0407a2e8c58d1e216

Edit routes

Edit routes: rtb-0407a2e8c58d1e216 / my-private-rt-ap-south-1c

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	CreateRouteTable
0.0.0.0/0	NAT Gateway	-	No	CreateRoute

Add route

nat-005c7e0bc4aadddc4

Use: "nat-005c7e0bc4aadddc4"

nat-0e6e00971ac7d34b1 (my-nat-ap-south-1b)

nat-00ce650d044b5eda3 (my-nat-ap-south-1a)

nat-01237f5c4bf7b7724 (my-nat-ap-south-1b)

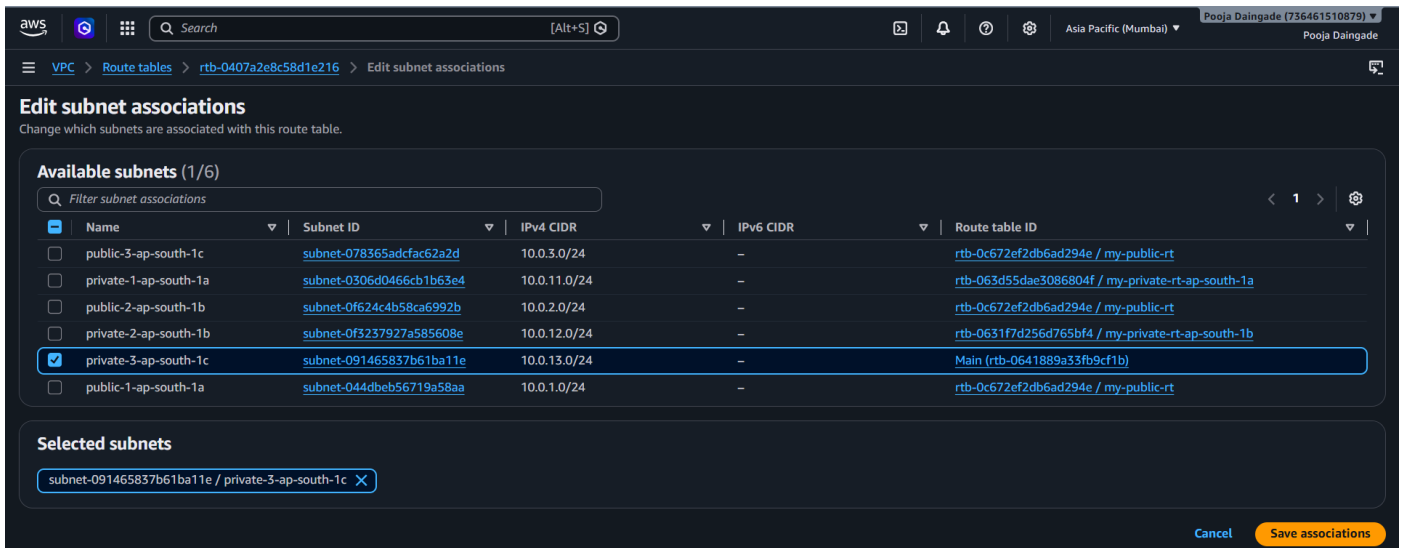
nat-005c7e0bc4aadddc4 (my-nat-ap-south-1c)

Cancel

Preview

Save changes

Associate Subnet: private-3-ap-south-1c



Edit subnet associations
Change which subnets are associated with this route table.

Available subnets (1/6)

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☐ public-3-ap-south-1c	subnet-078365adc62a2d	10.0.3.0/24	-	rtb-0c672ef2db6ad294e / my-public-rt
☐ private-1-ap-south-1a	subnet-0306d0466cb1b63e4	10.0.11.0/24	-	rtb-063d55dae3086804f / my-private-rt-ap-south-1a
☐ public-2-ap-south-1b	subnet-0f624c4b58ca6992b	10.0.2.0/24	-	rtb-0c672ef2db6ad294e / my-public-rt
☐ private-2-ap-south-1b	subnet-0f3237927a585608e	10.0.12.0/24	-	rtb-0631f7d256d765bf4 / my-private-rt-ap-south-1b
☒ private-3-ap-south-1c	subnet-091465837b61ba11e	10.0.13.0/24	-	Main (rtb-0641889a33fb9cf1b)
☐ public-1-ap-south-1a	subnet-044dbeb56719a58aa	10.0.1.0/24	-	rtb-0c672ef2db6ad294e / my-public-rt

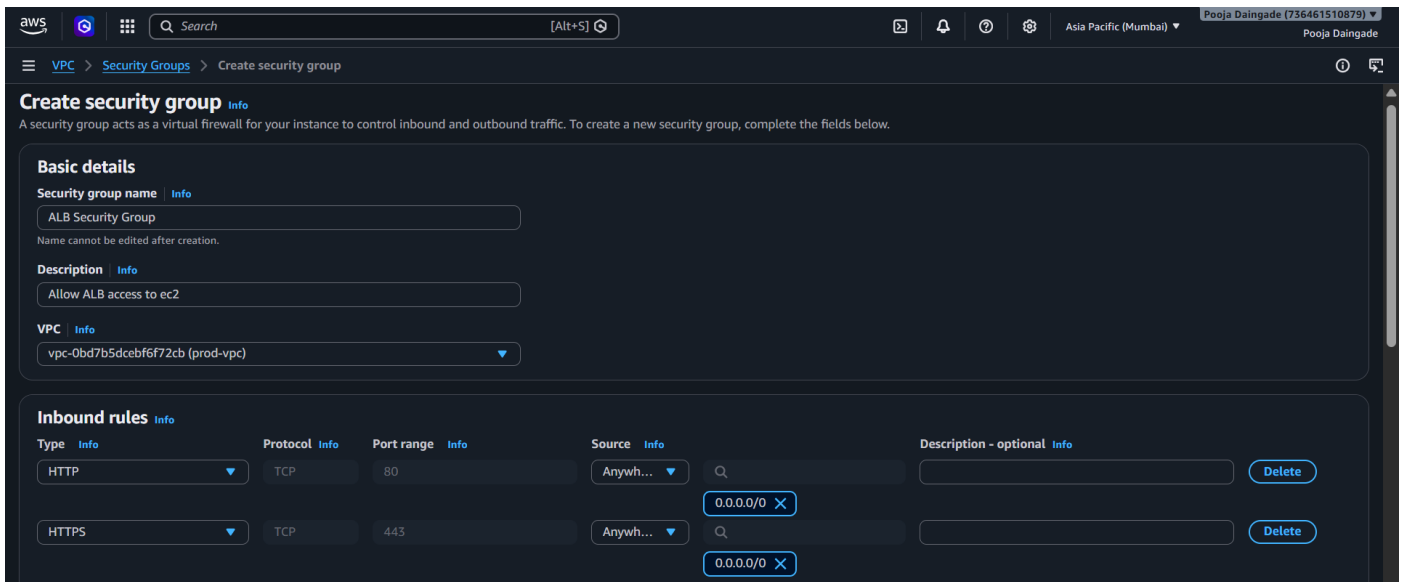
Selected subnets

subnet-091465837b61ba11e / private-3-ap-south-1c

Cancel Save associations

10. Create Security Group :

- Name: ALB security group
- Description: Allow ALB access to ec2
- VPC: prod-vpc
- Inbound rule: HTTP (80) → 0.0.0.0/0
HTTPS(443)→ 0.0.0.0/0



Create security group Info
A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name Info
ALB Security Group
Name cannot be edited after creation.

Description Info
Allow ALB access to ec2

VPC Info
vpc-0bd7b5dcebf6f72cb (prod-vpc)

Inbound rules Info

Type	Protocol	Port range	Source	Description - optional
HTTP	TCP	80	Anywh... 0.0.0.0/0	
HTTPS	TCP	443	Anywh... 0.0.0.0/0	

11. Create Security Group :

- Name: EC2 Security Group
- Description: Allow ec2-traffic
- VPC: prod-vpc
- Inbound rule: HTTP (80) → 0.0.0.0/0
HTTPS(443)→ 0.0.0.0/0

aws

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Pooja Daingade

VPC

Security Groups

Create security group

Create security group

Info

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name

Info

EC2 Security Group

Name cannot be edited after creation.

Description

Info

Allow ec2-traffic

VPC

Info

vpc-0bd7b5dceb6f6f72cb (prod-vpc)

Inbound rules

Info

Type

Info

Protocol

Info

Port range

Info

Source

Info

Description - optional

Info

HTTP

TCP

80

Anywh...

0.0.0.0/0

Delete

HTTPS

TCP

443

Anywh...

0.0.0.0/0

Delete

12.Launch Template:

- Name: web-template
- Version: v1
- AMI: Ubuntu
- Instance type: t3.micro
- AZ: ap-south-1a
- Sg: EC2-Security Group

aws

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EC2

Launch templates

Create launch template

Create launch template

Info

Creating a launch template allows you to create a saved instance configuration that can be reused, shared and launched at a later time. Templates can have multiple versions.

Launch template name and description

Launch template name - *required*

web-template

Must be unique to this account. Max 128 chars. No spaces or special characters like '&', "'", '@'.

Template version description

v1

Max 255 chars

Auto Scaling guidance

Info

Select this if you intend to use this template with EC2 Auto Scaling

☐ Provide guidance to help me set up a template that I can use with EC2 Auto Scaling

Summary

Software Image (AMI)

-

Virtual server type (instance type)

-

Firewall (security group)

-

Storage (volumes)

-

Cancel

Create launch template

EC2

Launch templates

Create launch template

Amazon Machine Image (AMI)

Ubuntu Server 26.04 LTS (HVM), SSD Volume Type

Free tier eligible

ami-01a00762f46d584a1 (64-bit (x86)) / ami-0b40571b9c2387b15 (64-bit (Arm))

Virtualization: hvm ENA enabled: true Root device type: ebs

Description

Ubuntu Server 26.04 LTS (HVM),EBS General Purpose (SSD) Volume Type. Support available from Canonical (<http://www.ubuntu.com/cloud/services>).

Canonical, Ubuntu, 26.04, amd64 resolute image

Architecture

AMI ID

Creation Date

Username

Verified provider

64-bit (x86)

ami-01a00762f46d584a1

2026-06-04

ubuntu

Instance type

Info

 |

Get advice

Instance type

t3.micro

Family: t3 2 vCPU 1 GiB Memory Current generation: true On-Demand Linux base pricing: 0.0112 USD per Hour Free tier eligible

On-Demand SUSE base pricing: 0.0112 USD per Hour On-Demand Windows base pricing: 0.0204 USD per Hour

On-Demand Ubuntu Pro base pricing: 0.0147 USD per Hour On-Demand RHEL base pricing: 0.04 USD per Hour

All generations

Compare instance types

Additional costs apply for AMIs with pre-installed software

Summary

Software Image (AMI)

Canonical, Ubuntu, 26.04, amd64...[read more](#)

ami-01a00762f46d584a1

Virtual server type (instance type)

t3.micro

Firewall (security group)

-

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Create launch template

User Data :

```
#!/bin/bash
```

```
apt update -y
```

```
apt install apache2 -y
```

```
systemctl enable apache2
```

```
systemctl start apache2
```

```
echo "Server: $(hostname)" > /var/www/html/index.html
```

13. Create Target Group:

- Target Type: Instances
- Target Group Name: web-tg
- VPC: prod-vpc
- Healthy threshold: 2
- Unhealthy threshold: 3
- Interval: 30s

aws

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EC2 > Target groups > Create target group

Step 1

Create target group

Step 2 - recommended

Register targets

Step 3

Review and create

Create target group

A target group can be made up of one or more targets. Your load balancer routes requests to the targets in a target group and performs health checks on the targets.

Settings - immutable

Choose a target type and the load balancer and listener will route traffic to your target. These settings can't be modified after target group creation.

Target type

Indicate what resource type you want to target. Only the selected resource type can be registered to this target group.

Instances

Supports load balancing to instances in a VPC. Integrate with Auto Scaling Groups or ECS services for automatic management.

Suitable for: ALB NLB GWLB

IP addresses

Supports load balancing to VPC and on-premises resources. Facilitates routing to IP addresses and network interfaces on the same instance. Supports IPv6 targets.

Suitable for: ALB NLB GWLB

Lambda function

Supports load balancing to a single Lambda function. ALB required as traffic source.

Suitable for: ALB

Application Load Balancer

Allows use of static IP addresses and PrivateLink with an Application Load Balancer. NLB required as traffic source.

Suitable for: NLB

Target group name

Name must be unique per Region per AWS account.

web-tg

Accepts: a-z, A-Z, 0-9, and hyphen (-). Can't begin or end with hyphen. 1-32 total characters; Count: 6/32

Protocol

Protocol for communication between the load balancer and targets.

Port

Port number where targets receive traffic. Can be overridden for individual targets.

EC2 > Target groups > Create target group

VPC

Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.

vpc-0bd7b5dceb6f72cb (prod-vpc)
10.0.0.0/16

Create VPC

Protocol version

HTTP1

Send requests to targets using HTTP/1.1. Supported when the request protocol is HTTP/1.1 or HTTP/2.

HTTP2

Send requests to targets using HTTP/2. Supported when the request protocol is HTTP/2 or gRPC, but gRPC-specific features are not available.

gRPC

Send requests to targets using gRPC. Supported when the request protocol is gRPC.

Health checks

The associated load balancer periodically sends requests, per the settings below, to the registered targets to test their status.

Health check protocol

HTTP

Health check path

Use the default path of "/" to perform health checks on the root, or specify a custom path if preferred.

/

Up to 1024 characters allowed.

EC2 > Target groups > Create target group

Advanced health check settings

Restore defaults

Health check port

The port the load balancer uses when performing health checks on targets. By default, the health check port is the same as the target group's traffic port. However, you can specify a different port as an override.

Traffic port

Override

Healthy threshold

The number of consecutive health checks successes required before considering an unhealthy target healthy.

2

2-10

Unhealthy threshold

The number of consecutive health check failures required before considering a target unhealthy.

3

2-10

Timeout

The amount of time, in seconds, during which no response means a failed health check.

5

seconds

2-120

Interval

The approximate amount of time between health checks of an individual target

30

seconds

14. Load Balancer:

- Type: Application Load Balancer
- Name: web-ALB
- Scheme: Internet-facing
- VPC: prod-vpc

Basic configuration

Load balancer name
Name must be unique within your AWS account and can't be changed after the load balancer is created.
web-ALB
A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme Info
Scheme can't be changed after the load balancer is created.

☒ **Internet-facing**
• Serves internet-facing traffic.
• Has public IP addresses.
• DNS name resolves to public IPs.
• Requires a public subnet.

☐ **Internal**
• Serves internal traffic.
• Has private IP addresses.
• DNS name resolves to private IPs.
• Compatible with the IPv4 and Dualstack IP address types.

Load balancer IP address type Info
Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address types. Public IPv4 addresses have an additional cost.

☒ **IPv4**
Includes only IPv4 addresses.

☐ **Dualstack**
Includes IPv4 and IPv6 addresses.

☐ **Dualstack without public IPv4**
Includes a public IPv6 address, and private IPv4 and IPv6 addresses. Compatible with internet-facing load balancers only.

Network mapping Info
The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.

VPC Info
The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to Lambda or on-premises targets, or if using VPC peering. To confirm the VPC for your targets, view [target groups](#).

vpc-0bd7b5dceb6f72cb (prod-vpc)
10.0.0.0/16

[Create VPC](#)

- Availability Zones and subnets:
 - ✓ ap-south-1a → public-1-ap-south-1a
 - ✓ ap-south-1b → public-2-ap-south-1b
 - ✓ ap-south-1c → public-3-ap-south-1c

Availability Zones and subnets Info
Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to traffic. The load balancer routes traffic to targets in the selected Availability Zones only.

☒ **ap-south-1a (aps1-az1)**

Subnet
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently. Selected subnets must have Network ACL inbound and outbound ALLOW rules, and internet gateway (IGW) routes configured to allow traffic.

subnet-044db5eb56719a58aa (public-1-ap-south-1a)
10.0.1.0/24
Network ACL ALLOW rules: Inbound (1) Outbound (1) | Route table: IGW (✓)

☒ **ap-south-1b (aps1-az3)**

Subnet
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently. Selected subnets must have Network ACL inbound and outbound ALLOW rules, and internet gateway (IGW) routes configured to allow traffic.

subnet-0f624c4b58ca6992b (public-2-ap-south-1b)
10.0.2.0/24
Network ACL ALLOW rules: Inbound (1) Outbound (1) | Route table: IGW (✓)

☒ **ap-south-1c (aps1-az2)**

Subnet
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently. Selected subnets must have Network ACL inbound and outbound ALLOW rules, and internet gateway (IGW) routes configured to allow traffic.

subnet-078365adcfac62a2d (public-3-ap-south-1c)
10.0.3.0/24
Network ACL ALLOW rules: Inbound (1) Outbound (1) | Route table: IGW (✓)

- Security Group: ALB Security Group
- Target Group: web-tg

EC2 > Load balancers > Create Application Load Balancer

Security groups

Select up to 5 security groups

ALB Security Group (sg-03695b264d88baaea) X

Inbound rule (2) Outbound rule (1)

Listeners and routing [Info](#)

A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.

▼ Listener HTTP:80 [Remove](#)

Protocol HTTP Port 80 1-65535

Default action [Info](#)

The default action is used if no other rules apply. Choose the default action for traffic on this listener.

Routing action

☒ Forward to target groups ☐ Redirect to URL ☐ Return fixed response

Forward to target group [Info](#)

Choose a target group and specify routing weight or [create target group](#).

Target group	Weight	Percent
web-tg Target type: Instance, IPv4 Target stickiness: Off	1 0-999	100%

15. Create Auto Scaling Group:

- Name: web-asg
- Template: web-template
- Version: 1

aws [EC2](#) > Auto Scaling groups > Create Auto Scaling group

Search [Alt+S]

Asia Pacific (Mumbai) Pooja Daingade (736461510879) Pooja Daingade

Step 1: Choose launch template

Choose launch template [Info](#)

Specify a launch template that contains settings common to all EC2 instances that are launched by this Auto Scaling group.

Name

Auto Scaling group name

Enter a name to identify the group.

web-asg

Must be unique to this account in the current Region and no more than 255 characters.

Launch template [Info](#)

For accounts created after May 31, 2023, the EC2 console only supports creating Auto Scaling groups with launch templates. Creating Auto Scaling groups with launch configurations is not recommended but still available via the CLI and API until December 31, 2023.

Launch template

Choose a launch template that contains the instance-level settings, such as the Amazon Machine Image (AMI), instance type, key pair, and security groups.

web-template [Create a launch template](#)

Version

1 [Create a launch template version](#)

- VPC: prod-vpc
- Availability Zones and subnets:
 - ✓ ap-south-1a → private-1-ap-south-1a
 - ✓ ap-south-1b → private-2-ap-south-1b
 - ✓ ap-south-1c → private-3-ap-south-1c

EC2 > Auto Scaling groups > Create Auto Scaling group

VPC
Choose the VPC that defines the virtual network for your Auto Scaling group.

vpc-074dcae9553661e11 (prod-vpc)
10.0.0.0/16

Create a VPC

Availability Zones and subnets
Define which Availability Zones and subnets your Auto Scaling group can use in the chosen VPC.

Select Availability Zones and subnets

aps1-az1 (ap-south-1a) | subnet-0c46c97468fbd5c56
(private-1-ap-south-1a)
10.0.11.0/24

aps1-az2 (ap-south-1c) | subnet-0051d2ab86c2b700f
(private-3-ap-south-1c)
10.0.13.0/24

aps1-az3 (ap-south-1b) | subnet-0ea1185a9a5c88711
(private-2-ap-south-1b)
10.0.12.0/24

Create a subnet

- Load balancing: Attach to an existing load balancer
- Target group: web-tg

EC2 > Auto Scaling groups > Create Auto Scaling group

Step 1: Choose launch template
Step 2: Choose instance launch options
Step 3 - optional: **Integrate with other services**
Step 4 - optional: Configure group size and scaling
Step 5 - optional: Add notifications
Step 6 - optional: Add tags
Step 7: Review

Integrate with other services - optional Info
Use a load balancer to distribute network traffic across multiple servers. Enable service-to-service communications with VPC Lattice. Shift resources away from impaired Availability Zones with zonal shift. You can also customize health check replacements and monitoring.

Load balancing Info
Use the options below to attach your Auto Scaling group to an existing load balancer, or to a new load balancer that you define.

Select Load balancing options

☐ No load balancer
Traffic to your Auto Scaling group will not be fronted by a load balancer.

☒ **Attach to an existing load balancer**
Choose from your existing load balancers.

☐ Attach to a new load balancer
Quickly create a basic load balancer to attach to your Auto Scaling group.

Attach to an existing load balancer

Select the load balancers to attach

☒ **Choose from your load balancer target groups**
This option allows you to attach Application, Network, or Gateway Load Balancers.

☐ Choose from Classic Load Balancers

Existing load balancer target groups
Only instance target groups that belong to the same VPC as your Auto Scaling group are available for selection.

Select target groups

web-tg | HTTP
Application Load Balancer: web-ALB

- Desired capacity: 3
- Min desired capacity: 1
- Max desired capacity: 7

Step 1: Choose launch template
Step 2: Choose instance launch options
Step 3 - optional: Integrate with other services
Step 4 - optional: **Configure group size and scaling**
Step 5 - optional: Add notifications
Step 6 - optional: Add tags
Step 7: Review

Configure group size and scaling - optional Info
Define your group's desired capacity and scaling limits. You can optionally add automatic scaling to adjust the size of your group.

Group size Info
Set the initial size of the Auto Scaling group. After creating the group, you can change its size to meet demand, either manually or by using automatic scaling.

Desired capacity type
Choose the unit of measurement for the desired capacity value. vCPUs and Memory(GiB) are only supported for mixed instances groups configured with a set of instance attributes.

Units (number of instances)

Desired capacity
Specify your group size.
3

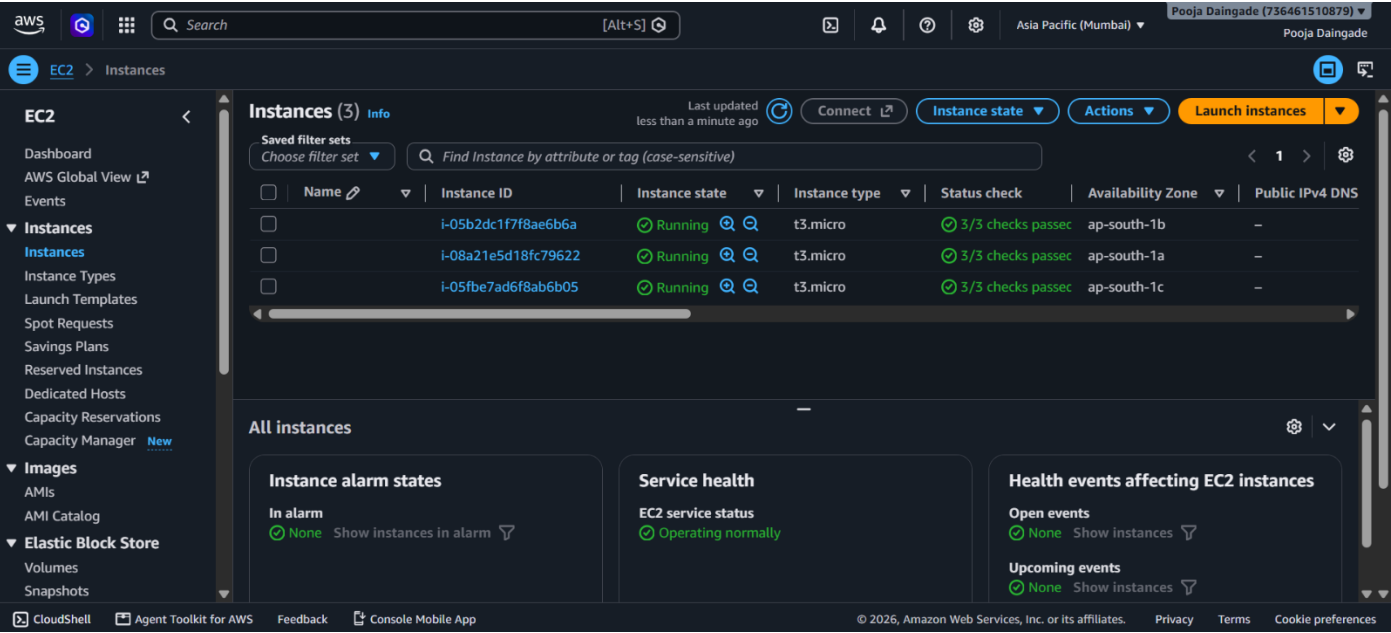
Scaling Info
You can resize your Auto Scaling group manually or automatically to meet changes in demand.

Scaling limits
Set limits on how much your desired capacity can be increased or decreased.

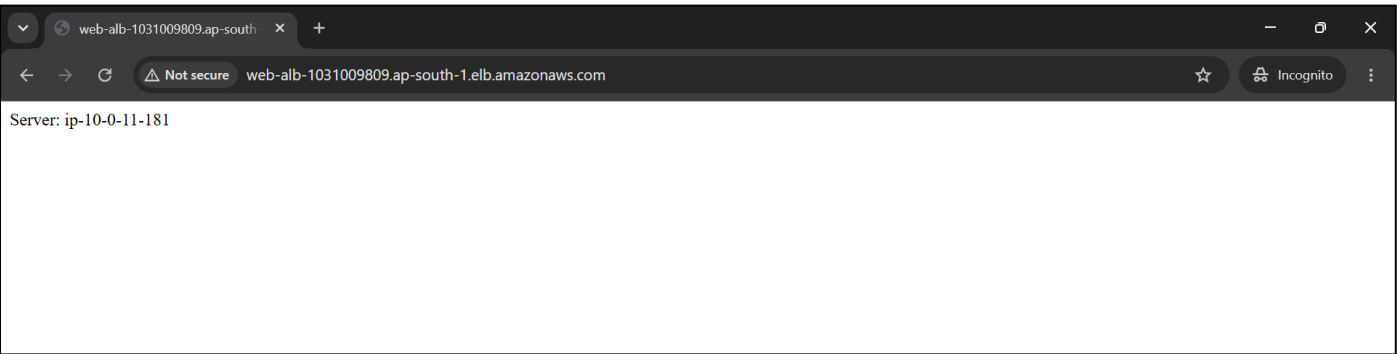
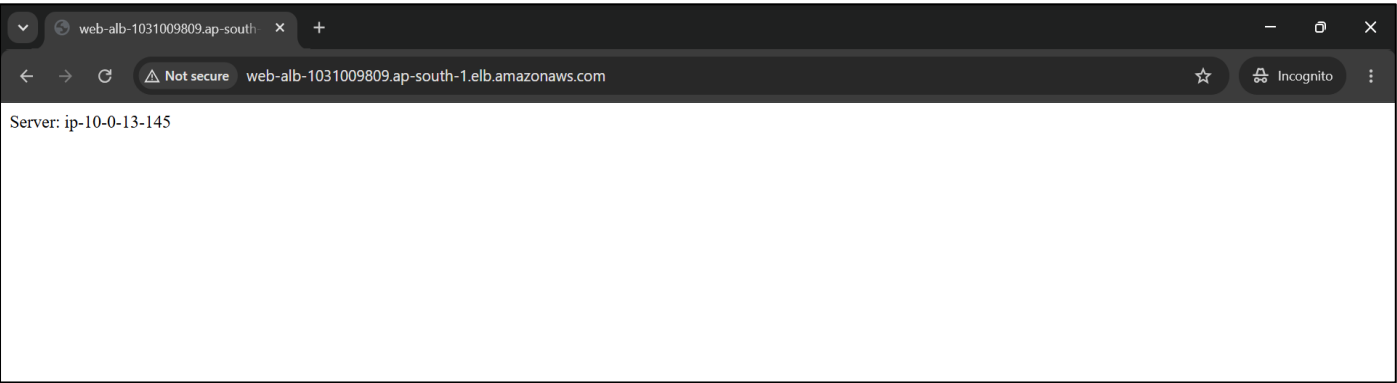
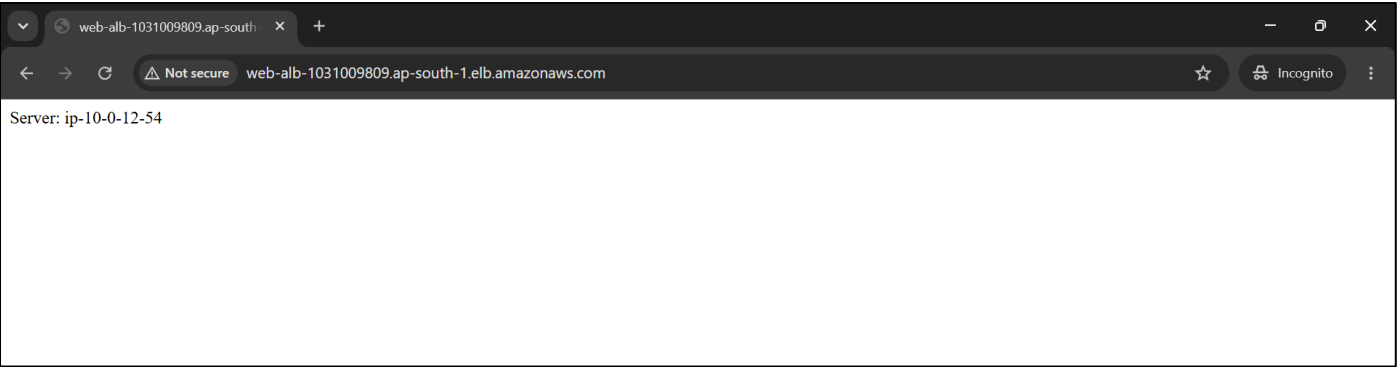
Min desired capacity
1
Equal or less than desired capacity

Max desired capacity
7
Equal or greater than desired capacity

16. Three EC2 instances automatically launched by the Auto Scaling Group.



17. Testing: Access ALB DNS. Refresh page → different server hostname



18. Requested a public SSL/TLS certificate in **AWS Certificate Manager (ACM)**.

- Domain name: *.poojadaingade.shop

Request public certificate

Domain names
Provide one or more domain names for your certificate.

Fully qualified domain name [Info](#)

*.poojadaingade.shop

[Add another name to this certificate](#)

You can add additional names to this certificate. For example, if you're requesting a certificate for "www.example.com", you might want to add the name "example.com" so that customers can reach your site by either name.

Allow export [Info](#)

☒ **Disable export**
Use this certificate only with integrated AWS services. The private key for this certificate will be disallowed for exporting from AWS.

☐ **Enable export**
Export this certificate and private key for use with any TLS workflow. ACM will charge your account based on the requested domains when the certificate is issued for the first time and for each renewal.

- Create DNS record in Route 53 → Create record → CNAME record is created in Route 53 → Certificate is issued

Create DNS records in Amazon Route 53 (1/1)

Search domains 1 match

Validation status = Pending validation or Validation status = Failed and Is domain in Route 53? = Yes [Clear filters](#)

☒ Domain	☒ Validation status	☒ Is domain in Route 53?
*.poojadaingade.shop	Pending validation	Yes

[Cancel](#) [Create records](#)

19. Open **web-ALB** load balancer → Listeners and rules → Add listeners

- Protocol- HTTPS
- Target group: web-tg

Listener: HTTPS:443
A listener checks for connection requests on its configured protocol and port. Incoming requests are evaluated against rules in priority order; the first matching rule determines routing. If no rule matches, the default action is applied.

Protocol
Used for connections from clients to the load balancer.
HTTPS

Port
The port on which the load balancer is listening for connections.
443

Default action [Info](#)
The default action is used if no other rules apply. Choose the default action for traffic on this listener.

Pre-routing action [Info](#)

☒ No pre-routing action ☐ Authenticate user ☐ Validate token

Routing action

☒ Forward to target groups ☐ Redirect to URL ☐ Return fixed response

Forward to target group [Info](#)
Choose a target group and specify routing weight or [create target group](#).

Target group	Weight	Percent
web-tg Target type: Instance, IPv4 Target stickiness: Off	1	100%

- Select ACM certificate → *.poojadaingade.shop

Secure listener settings

Security policy | Info
Your load balancer uses a Secure Socket Layer (SSL) negotiation configuration called a security policy to manage SSL connections with clients. [Compare security policies](#)

Security category | Policy name | New
All security policies | ELBSecurityPolicy-TLS13-1-2-Res-PQ-2025-09 (Recommended)

Default SSL/TLS server certificate
The certificate used if a client connects without SNI protocol, or if there are no matching certificates. You can source this certificate from AWS Certificate Manager (ACM), Amazon Identity and Access Management (IAM), or import a certificate. This certificate will automatically be added to your listener certificate list.

Certificate source
☒ From ACM
☐ From IAM
☐ Import certificate

Certificate (from ACM)
The selected certificate will be applied as the default SSL/TLS server certificate for this load balancer's secure listeners.
 *.poojadaingade.shop
 64516f52-d19d-46d5-87f8-afb30731c609
[Request new ACM certificate](#)

Client certificate handling | Info
Client certificates are used to make authenticated requests to remote servers. [Learn more](#)

☐ **Mutual authentication (mTLS)**
Mutual TLS (Transport Layer Security) authentication offers two-way peer authentication. It adds a layer of security over TLS and allows your services to verify the client that's making the connection.

Default attributes are applied when adding a listener. You can edit them after adding the listener.

Listener tags - optional

20. Route 53 → Hosted zone → poojadaingade.shop → Create record

- Record name: api
- Record type: A
- Alias: On
- Choose endpoint: Alias to Application and Classic Load Balancer
- Choose region: Asia Pacific (Mumbai)
- Select Load Balancer

Create record | Info

Quick create record | [Switch to wizard](#)

Record 1 | [Delete](#)

Record name | Info
api .poojadaingade.shop
Keep blank to create a record for the root domain.

Record type | Info
A - Routes traffic to an IPv4 address and some AWS resources

☒ **Alias**

Route traffic to | Info
Alias to Application and Classic Load Balancer
Asia Pacific (Mumbai)
Q dualstack.web-ALB-1031009809.ap-south-1.elb.amazonaws.com
Alias hosted zone ID: ZP97RAFLXTNZK

Routing policy | Info
Simple routing

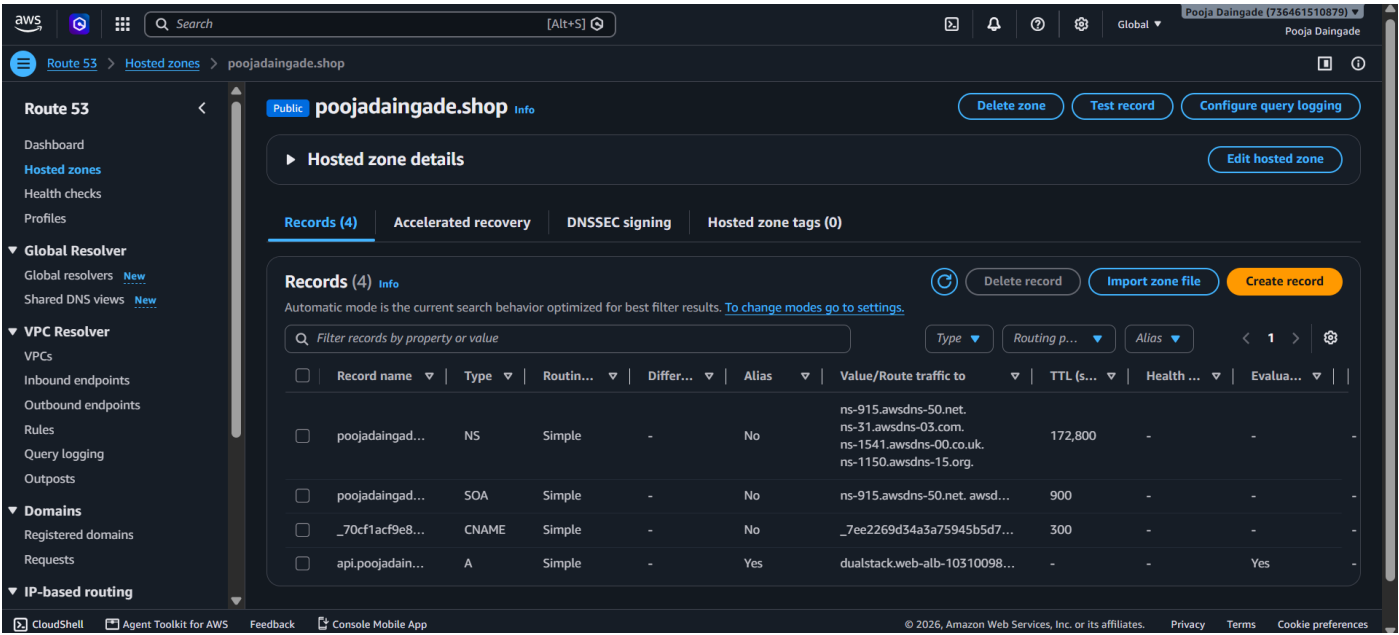
Evaluate target health
☒ Yes

[Add another record](#)

[Cancel](#) [Create records](#)

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21. A record configured in Route 53 to route api.poojadaingade.shop traffic to the ALB.



22. Accessed the domain api.poojadaingade.shop and successfully verified the server response.

